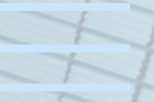




LENI GUSTIA



SUPERSTORE SALES AND PROFIT ANALYSIS

IDENTIFYING KEY BUSINESS INSIGHTS
USING SQL



PROJECT BACKGROUND

In today's competitive retail industry, understanding sales performance and profitability is essential for making data-driven business decisions. This project analyzes the **Superstore dataset** to identify key insights related to product performance, customer segments, regional trends, and the impact of discounts. **Using SQL for data exploration and Power BI for interactive dashboard reporting**, this project aims to support strategic business improvement through actionable insights.

BUSINESS PROBLEM



The Superstore company faces increasing market competition and needs to better understand its business performance. Several important questions need to be addressed, including:

- Which product categories and sub-categories generate the highest sales and profit?
- Which products or segments contribute to financial losses?
- Which regions perform best in terms of sales and profitability?
- Which customer segments generate the most revenue?
- How does discount affect profitability?
- What data-driven strategies can be recommended to improve profitability and operational efficiency?

DATASET INFORMATION



INFORMATION

- ✔ Dataset Name : Sample Superstore Dataset
- ✔ Source : [Kaggle](#)
- ✔ Type : Retail transactional data
- ✔ Number of Records: 9,988 rows (after data cleaning)
- ✔ Number of Features: 21 columns

MAIN VARIABLE

- Order information (Order ID, Order Date, Ship Date, Ship Mode)
- Customer information (Customer ID, Customer Name, Segment)
- Geographic information (Country, City, State, Region)
- Product information (Category, Sub-Category, Product Name)
- Sales metrics (Sales, Quantity, Discount, Profit)

TOOLS USED



Initial data inspection and validation



Data cleaning and SQL analysis



Dashboard visualization and storytelling



Data preprocessing

Raw dataset

There are 9,994 rows and 21 columns



Convert date format

Transform order_date and ship_date from text into DATE format for time-based analysis



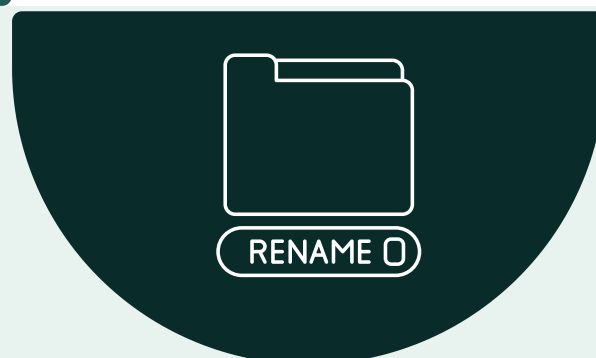
Identify incomplete records

Detected 6 rows with missing values in critical columns (sales, quantity, discount, profit)



Rename table & standardize column names

Convert column names into snake_case format for consistency



Data quality checking

Validate total records, duplicate values, and missing data



Raw dataset

Data Superstore																				Search		Comments											
File Home Insert Page Layout Formulas Data Review View Help Nitro Pro Acrobat																																	
Clipboard		Font		Paragraph		Alignment		Number		Styles		Conditional Formatting		Format as Table		Cell Styles		Insert		Delete		Format		AutoSum		Filter		Sort & Filter		Find & Select		Add-Ins	
A1		v		x		fx		Row ID																									
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U													
Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit														
1	CA-2016-1	11/08/2016	11/11/2016	Second	Cl CG-12520	Claire Gut Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-BD-31	Furniture	Bookcases	Bush Some	261.96	2	0	41.9130														
2	CA-2016-1	11/08/2016	11/11/2016	Second	Cl CG-12520	Claire Gut Consumer	United Sta	Henderson	Kentucky	42420	South	FUR-CH-31	Furniture	Chairs	Hon Delux	721.94	3	0	219.582														
3	CA-2016-1	06/12/2016	6/16/2016	Second	Cl DV-13045	Darrin Van Corporate	United Sta	Los Angeles	California	90036	West	OFF-LA-10	Office Sup	Labels	Self-Adhes	14.62	2	0	6.8714														
4	US-2015-1	10/11/2015	10/18/2015	Standard	CSO-20335	Sean O'Do Consumer	United Sta	Fort Laude	Florida	33311	South	FUR-TA-31	Furniture	Tables	Bretford C	957.5775	5	0.45	-383.031														
5	US-2015-1	10/11/2015	10/18/2015	Standard	CSO-20335	Sean O'Do Consumer	United Sta	Fort Laude	Florida	33311	South	OFF-ST-10	Office Sup	Storage	Eldon Fold	22.368	2	0.2	2.5164														
6	CA-2014-1	06/09/2014	6/14/2014	Standard	CBH-11710	Brosina H Consumer	United Sta	Los Angeles	California	90032	West	FUR-FU-31	Furniture	Furnishings	Eldon Expr	68.86	7	0	14.1694														
7	CA-2014-1	06/09/2014	6/14/2014	Standard	CBH-11710	Brosina H Consumer	United Sta	Los Angeles	California	90032	West	OFF-AR-10	Office Sup	Art	Newell 32	7.28	4	0	1.9650														
8	CA-2014-1	06/09/2014	6/14/2014	Standard	CBH-11710	Brosina H Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-10	Technology	Phones	Mitel 5320	907.152	6	0.2	90.7152														
9	CA-2014-1	06/09/2014	6/14/2014	Standard	CBH-11710	Brosina H Consumer	United Sta	Los Angeles	California	90032	West	OFF-BI-10	Office Sup	Binders	Dixi Angle	18.504	3	0.2	5.7825														
10	CA-2014-1	06/09/2014	6/14/2014	Standard	CBH-11710	Brosina H Consumer	United Sta	Los Angeles	California	90032	West	OFF-AP-10	Office Sup	Appliances	Belkin F5C	114.9	5	0	34.45														
11	CA-2014-1	06/09/2014	6/14/2014	Standard	CBH-11710	Brosina H Consumer	United Sta	Los Angeles	California	90032	West	FUR-TA-31	Furniture	Tables	Chromel	1706.184	9	0.2	85.3092														
12	CA-2014-1	06/09/2014	6/14/2014	Standard	CBH-11710	Brosina H Consumer	United Sta	Los Angeles	California	90032	West	TEC-PH-10	Technology	Phones	Konstel 25	911.424	4	0.2	68.3564														
13	CA-2017-1	14/15/2017	4/20/2017	Standard	CAA-10480	Andrew Al Consumer	United Sta	Concord	North Car	28027	South	OFF-PA-10	Office Sup	Paper	Xerox 196	15.552	3	0.2	5.4432														
14	CA-2016-1	12/05/2016	12/10/2016	Standard	IM-15070	Irene Mad Consumer	United Sta	Seattle	Washington	98103	West	OFF-BI-10	Office Sup	Binders	Fellowes P	402.976	3	0.2	132.5922														
15	US-2015-1	11/22/2015	11/26/2015	Standard	CHP-14815	Harold Pav Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-AP-10	Office Sup	Appliances	Holmes Re	68.81	5	0.8	-123.858														
16	US-2015-1	11/22/2015	11/26/2015	Standard	CHP-14815	Harold Pav Home Offi	United Sta	Fort Worth	Texas	76106	Central	OFF-BI-10	Office Sup	Binders	Stores Dur	2.544	3	0.8	-3.816														
17	CA-2014-1	11/11/2014	11/18/2014	Standard	CFK-19075	Pete Kriz Consumer	United Sta	Madison	Wisconsin	53711	Central	OFF-ST-10	Office Sup	Storage	Stur-D Sto	605.88	6	0	13.3170														
18	CA-2014-1	5/13/2014	5/15/2014	Second	Cl AG-10270	Alejandro Consumer	United Sta	West Jord	Utah	84084	West	OFF-ST-10	Office Sup	Storage	Fellowes S	55.5	2	0	9.95														
19	CA-2014-1	8/27/2014	09/01/2014	Second	Cl ZD-21925	Zuschuss E Consumer	United Sta	San Francisco	California	94109	West	OFF-AR-10	Office Sup	Art	Newell 34	8.56	2	0	2.4824														
20	CA-2014-1	8/27/2014	09/01/2014	Second	Cl ZD-21925	Zuschuss E Consumer	United Sta	San Francisco	California	94109	West	TEC-PH-10	Technology	Phones	Cisco SPA	213.48	3	0.2	16.015														
21	CA-2014-1	8/27/2014	09/01/2014	Second	Cl ZD-21925	Zuschuss E Consumer	United Sta	San Francisco	California	94109	West	OFF-BI-10	Office Sup	Binders	Wilson Jor	22.72	4	0.2	7.384														
22	CA-2016-1	12/09/2016	12/13/2016	Standard	KB-16585	Ken Black Corporate	United Sta	Fremont	Nebraska	68025	Central	OFF-AR-10	Office Sup	Art	Newell 331	19.46	7	0	5.0590														
23	CA-2016-1	12/09/2016	12/13/2016	Standard	KB-16585	Ken Black Corporate	United Sta	Fremont	Nebraska	68025	Central	OFF-AP-10	Office Sup	Appliances	Acco Six-D	60.34	7	0	15.688														

Rename table & standardize column names

```
# Change the date column format from text to date
update data_superstore
set
    order_date = str_to_date(order_date, '%m/%d/%Y'),
    ship_date = str_to_date(ship_date, '%m/%d/%Y');
```

Converted date columns into MySQL DATE format to support accurate time-based analysis.



Convert date format

```
# Change column names to snake_case format
alter table data_superstore
change `Row ID` row_id int,
change `Order ID` order_id varchar(30),
change `Order Date` order_date varchar(20),
change `Ship Date` ship_date varchar(20),
change `Ship Mode` ship_mode varchar(50),
change `Customer ID` customer_id varchar(30),
change `Customer Name` customer_name varchar(100),
change `Postal Code` postal_code varchar(20),
change `Product ID` product_id varchar(50),
change `Sub-Category` sub_category varchar(50),
change `Product Name` product_name text;
```

Standardized the table and column names into snake_case format for better consistency and easier SQL querying.

	Field	Type	Null	Key	Default	Extra
▶	row_id	int	YES		NULL	
	order_id	varchar(30)	YES		NULL	
	order_date	date	YES		NULL	
	ship_date	date	YES		NULL	
	ship_mode	varchar(50)	YES		NULL	
	customer_id	varchar(30)	YES		NULL	
	customer_name	varchar(100)	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	postal_code	varchar(20)	YES		NULL	
	Region	text	YES		NULL	
	product_id	varchar(50)	YES		NULL	
	Category	text	YES		NULL	
	sub_category	varchar(50)	YES		NULL	
	product_name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	int	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

Clean and analysis-ready dataset after column standardization and date conversion.

Data quality checking

```
# Checking the distance and number of row_ids
select
  min(row_id) as min_row_id,
  max(row_id) as max_row_id,
  count(row_id) as total_row_id
from data_superstore;
```

	min_row_id	max_row_id	total_row_id
▶	1	9994	9988

Confirmed a gap between the expected and actual number of records, indicating missing entries in the dataset.

```
# Finding missing row_id
with recursive numbers as (
  select 1 as n
  union all
  select n + 1
  from numbers
  where n < 9994
)
select n as missing_row_id
from numbers
where n not in (
  select row_id
  from data_superstore
);
```

Identified 6 missing row IDs: 182, 431, 432, 1407, 1970, and 1972.

	missing_row_id
▶	182
	431
	432
	1407
	1970
	1972

```
# Check if there are any duplicate row_ids
select
  row_id,
  count(*) as duplicate_count
from data_superstore
group by row_id
having count(*) > 1;
```

row_id	duplicate_count
--------	-----------------

No duplicate records were found, confirming that each row_id is unique.

Final Dataset

Identify incomplete records

182	Tenex File Box, Personal Filing Tote with Lid, Black	24.816	2	0.2	1.8612
183	Imation Secure+ Hardware Encrypted USB 2.0 Flash Drive				
184	AT&T TR1909W	503.96	4	0	131.0296
185	Nokia Lumia 521 (T-Mobile)	149.95	5	0	41.986

Found incomplete records with missing values in key numerical columns (sales, quantity, discount, and profit), which were excluded from further analysis.

Country	City	State	postal_code	Region	product_id	Category	sub_category	product_name	Sales	Quantity	Discount	Profit
United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection Bookcase	261.96	2	0	41.9136
United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	731.94	3	0	219.582
United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Labels for Typewriters b...	14.62	2	0	6.8714
United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim Rectangular Table	957.5775	5	0.45	-383.031
United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold 'N Roll Cart System	22.368	2	0.2	2.5164
United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood and Plastic Desk Acces...	48.86	7	0	14.1694
United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322	7.28	4	0	1.9656
United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP phone	907.152	6	0.2	90.7152
United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders with Locking Rings by S...	18.504	3	0.2	5.7825
United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206VTEL 6 Outlet Surge	114.9	5	0	34.47
United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092
United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konftel 250 Conference phone - Charcoal black	911.424	4	0.2	68.3568
United States	Concord	North C...	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967	15.552	3	0.2	5.4432
United States	Seattle	Washing...	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic Comb Binding Machine	407.976	3	0.2	132.5922
United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacement Filter for HEPA Air Cleaner...	68.81	5	0.8	-123.858
United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders	Storex DuraTech Recycled Plastic Frosted Binders	2.544	3	0.8	-3.816
United States	Madison	Wisconsin	53711	Central	OFF-ST-10004186	Office Supplies	Storage	Stur-D-Stor Shelving, Vertical 5-Shelf: 72"H x 3...	665.88	6	0	13.3176
United States	West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Drawer	55.5	2	0	9.99
United States	San Francisco	California	94109	West	OFF-AR-10003056	Office Supplies	Art	Newell 341	8.56	2	0	2.4824
United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone	213.48	3	0.2	16.011
United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging View Binder, White, 1"	22.72	4	0.2	7.384
United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	Office Supplies	Art	Newell 318	19.46	7	0	5.0596
United States	Fremont	Nebraska	68025	Central	OFF-AP-10001492	Office Supplies	Appliances	Acron Six-Outlet Power Strip, 4' Cord Length	60.34	7	0	15.6884

Final cleaned dataset contains 9,988 rows and is ready for further analysis.



Exploratory Data Analysis



Q1. What is the company's total sales and total profit?

Syntax

```
Select sum(Sales) as total_sales,
sum(Profit) as total_profit from
data_superstore;
```

Results

	total_sales	total_profit
▶	2295273.9242999554	285988.27770000126

Insight

The company generated **strong total sales of 2.29 million** and maintained a **positive profit of 285.99 thousand**, indicating solid business performance and effective operational profitability.

Q2. Which product category generates the highest sales and profit?

Syntax

```
select category, sum(Sales) as total_sales,
sum(Profit) as total_profit
from data_superstore group by category
order by total_sales desc;
```

Results

	category	total_sales	total_profit
▶	Technology	834227.0969999966	145046.20409999986
	Furniture	741999.7952999998	18451.2728
	Office Supplies	719047.0320000029	122490.80080000011

Insight

The **Technology category** is the company's main growth driver, generating the highest sales and profit. Meanwhile, **Furniture** shows lower profitability despite strong sales, indicating the **need for margin improvement**, while **Office Supplies** provides stable and consistent profit.

Q3. Which sub-categories are the most profitable, and which are causing losses?

Syntax

```
select sub_category, sum(Profit) as
total_profit
from data_superstore
group by sub_category
order by total_profit desc;
```

Results

	sub_category	total_profit
▶	Copiers	55617.82490000001
	Phones	44515.7306
	Accessories	41527.89169999994
	Paper	34053.56929999997
	Binders	30221.763299999995
	Chairs	26590.166300000026
	Storage	21278.826399999998
	Appliances	18138.005399999995
	Furnishings	13059.143599999983
	Envelopes	6964.176700000003
	Art	6527.786999999998
	Labels	5546.253999999998
	Machines	3384.7569
	Fasteners	949.5181999999998
	Supplies	-1189.0994999999984
	Bookcases	-3472.5559999999978
	Tables	-17725.481100000008

Insight

Copiers, Phones, and Accessories are the most **profitable** sub-categories and should be prioritized for growth. In contrast, **Tables and Bookcases** generate **losses**, indicating potential pricing, discount, or cost management issues that require further evaluation.



Exploratory Data Analysis



Q4. Which region has the best sales and profit performance?

Syntax

```
select region, sum(Sales) as total_sales,
sum(Profit) as total_profit
from data_superstore
group by region
order by total_profit desc;
```

Results

	region	total_sales	total_profit
▶	West	725457.8245000006	108418.44890000018
	East	678664.4559999979	91500.88300000026
	South	391021.2010000004	46618.04830000007
	Central	500130.44280000054	39450.897499999985

Insight

The **West region** achieved the **highest sales and profit**, making it the strongest-performing market. Meanwhile, the **Central region** generated **relatively high sales but lower profit**, indicating potential opportunities to improve operational efficiency and profitability.

Q5. Which customer segment contributes the most to total sales?

Syntax

```
select segment, sum(Sales) as total_sales
from data_superstore
group by segment
order by total_sales desc;
```

Results

	segment	total_sales
▶	Consumer	1161109.3849999886
	Corporate	705737.6228
	Home Office	428426.9165000003

Insight

The **Consumer** segment contributes the **largest share of total sales**, making it the company's primary customer base. Meanwhile, **Corporate and Home Office** segments **present additional opportunities** for targeted growth and market expansion.

Q6. Who are the top customers with the highest total purchases?

Syntax

```
select Customer_Name, sum(Sales) as
total_purchase
from data_superstore
group by Customer_Name
order by total_purchase desc limit 10;
```

Results

	Customer_Name	total_purchase
▶	Sean Miller	25043.05
	Tamara Chand	19052.217999999997
	Raymond Buch	15117.339
	Tom Ashbrook	14595.62
	Adrian Barton	14473.570999999998
	Ken Lonsdale	14175.229
	Sanjit Chand	14142.333999999999
	Hunter Lopez	12873.297999999999
	Sanjit Engle	12209.438000000002
	Christopher Conant	12129.072

Insight

Sean Miller is the **highest-value customer**, contributing the largest total purchases. Identifying and retaining high-value customers like these is essential for strengthening customer loyalty and maximizing long-term revenue.



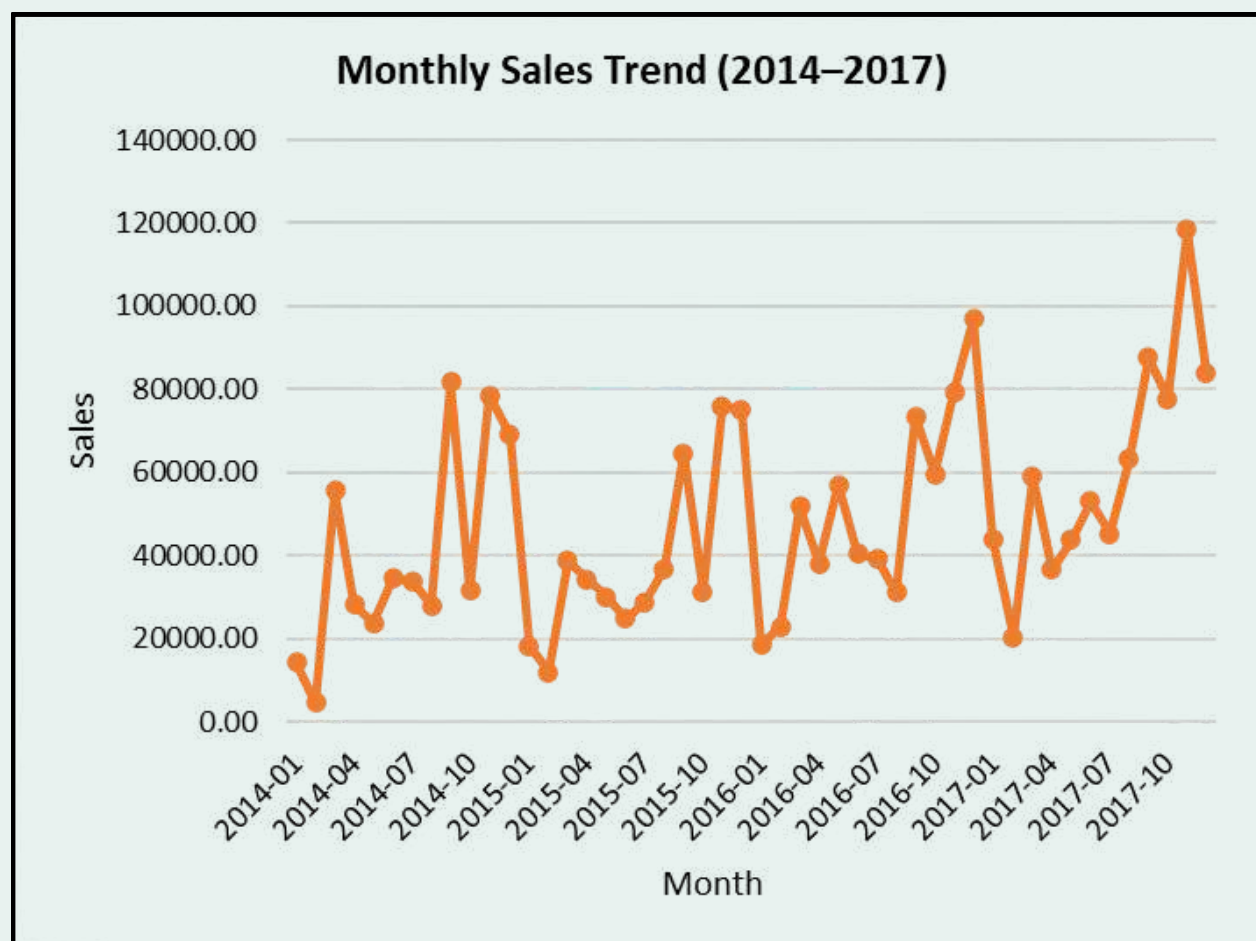
Exploratory Data Analysis

Q7. How do sales trends change over time?

Syntax

```
select date_format(order_date, '%Y-%m') as month, sum(sales) as total_sales
from data_superstore
group by month
order by month;
```

Results



Insight

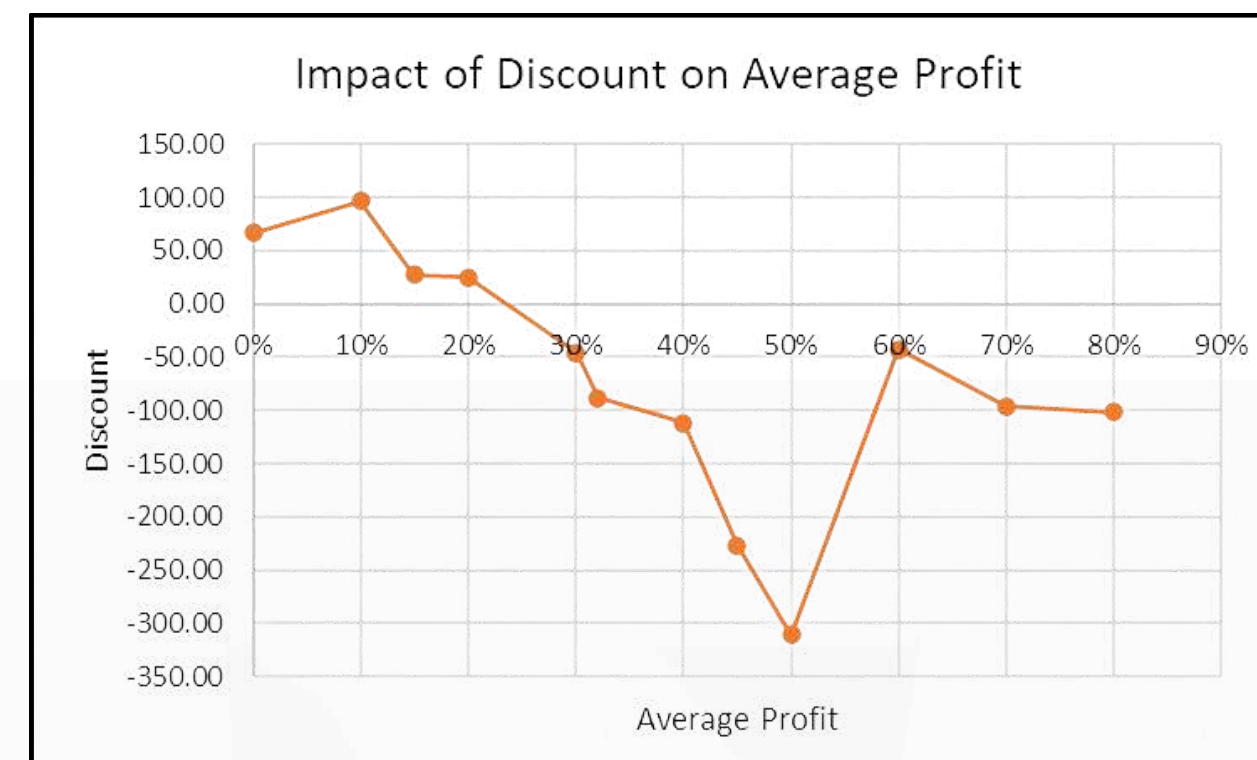
Sales show an overall upward trend from 2014 to 2017, indicating business growth over time. **Strong sales peaks** consistently occur toward the end of each year, especially in **November and December**, suggesting seasonal demand patterns.

Q8. Does the discount given affect the company's profit?

Syntax

```
select Discount, avg(Profit) as average_profit
from data_superstore
group by discount
order by discount;
```

Results



Insight

Higher discount rates are associated with lower profitability. While small discounts still generate positive profit, **discounts above 30%** consistently lead to losses, indicating that excessive discounting negatively impacts the company's profit margins.



Q9. Which products have high sales but generate low or negative profit?

Syntax

```
select product_name, sum(sales) as
total_sales, sum(profit) as total_profit
from data_superstore
group by product_name
having sum(profit) < 0
order by total_sales desc
limit 5;
```

Results

Result Grid			
Filter Rows:		Export:	Wrap Cell Content: Fetch rows
	product_name	total_sales	total_profit
▶	Cisco TelePresence System EX90 Videoconferen...	22638.48	-1811.0784
	GBC DocuBind P400 Electric Binding System	17965.068	-1878.1662000000003
	High Speed Automatic Electric Letter Opener	17030.311999999998	-262.00480000000005
	Lexmark MX611dhe Monochrome Laser Printer	16829.901	-4589.973
	Martin Yale Chadless Opener Electric Letter Ope...	16656.199999999997	-1299.1836

Insight

Several **top-selling products** are still generating losses, indicating that high sales volume does not always lead to profitability. This suggests potential issues such as excessive **discounting** or **inefficient pricing**, **highlighting the need to review pricing strategies** and product margins for these items.



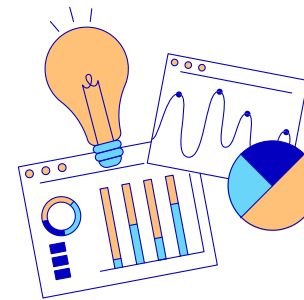
From Data to Dashboard



Taking SQL findings one step further —
into an interactive visual experience

PHASE 1 - SQL ANALYSIS

- ✓ Identified key profit drivers
- ✓ Detected loss-making products
- ✓ Analyzed regional performance
- ✓ Uncovered discount impact



PHASE 2 - POWER BI DASHBOARD

- Visualized all findings interactively
- Enabled dynamic filtering by Year, Region, Segment & Category
- Surfaced actionable business insights

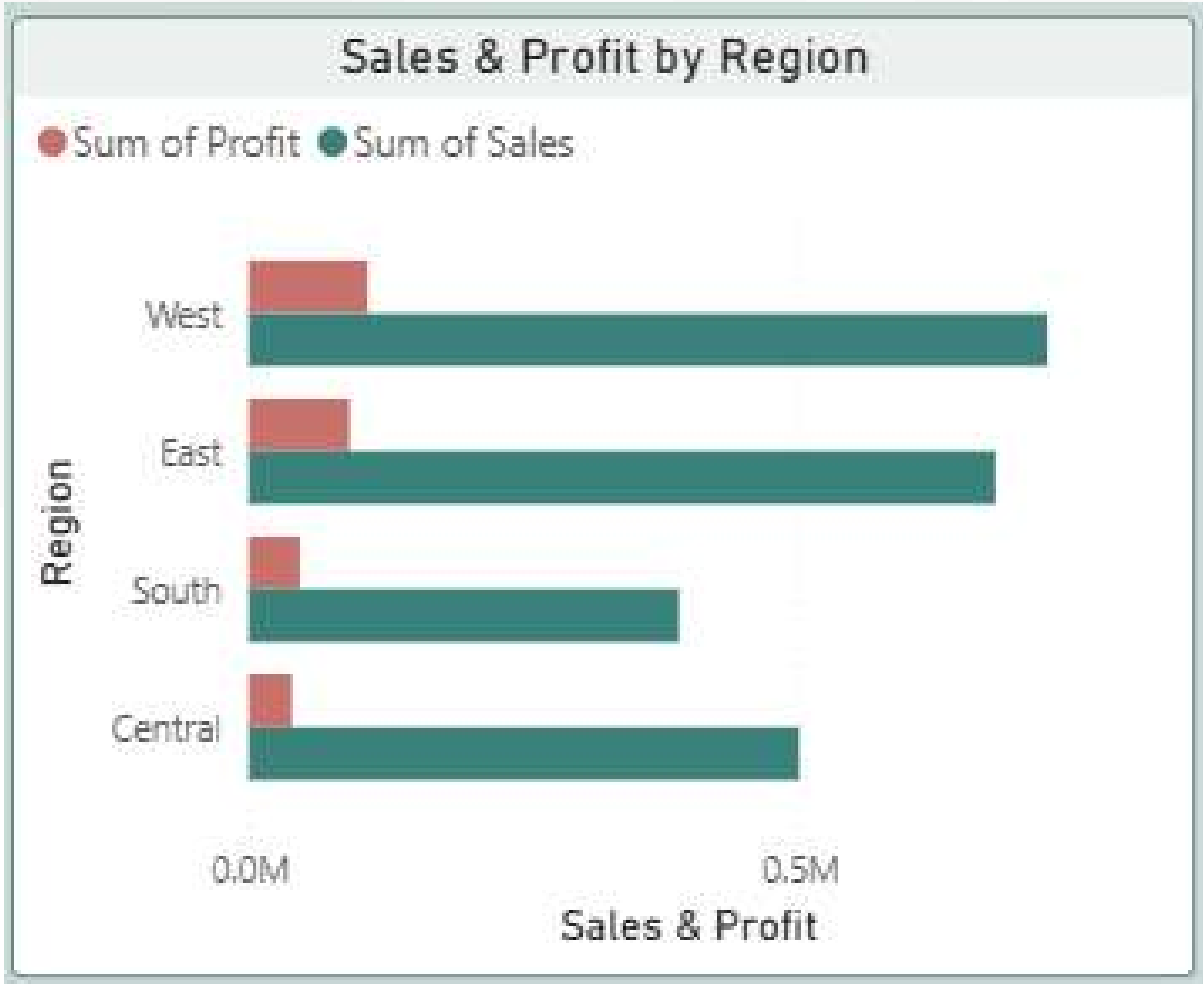


Interactive Sales & Profit Dashboard





Interactive Sales & Profit Dashboard



FINDING



Revenue grows steadily throughout the year, peaking significantly in **Q4 (Nov-Dec)**.

INSIGHT



Despite strong Q4 sales, **profit margins narrow** during the same period, indicating heavy discounting activity during the **holiday season**.

FINDING



West region leads in both sales and profit. **Central** has moderate sales but the lowest profit.

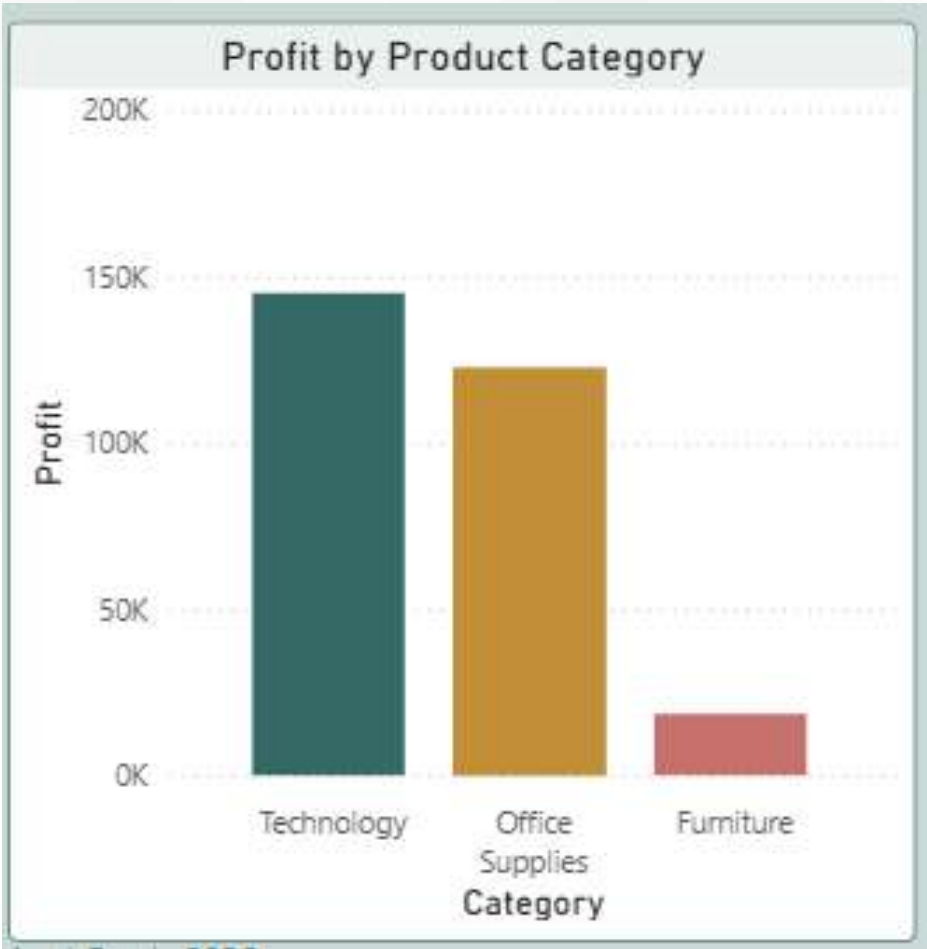
INSIGHT



Central is the only region operating at a loss, driven by the highest discount rate at 22%, significantly above other regions.



Interactive Sales & Profit Dashboard

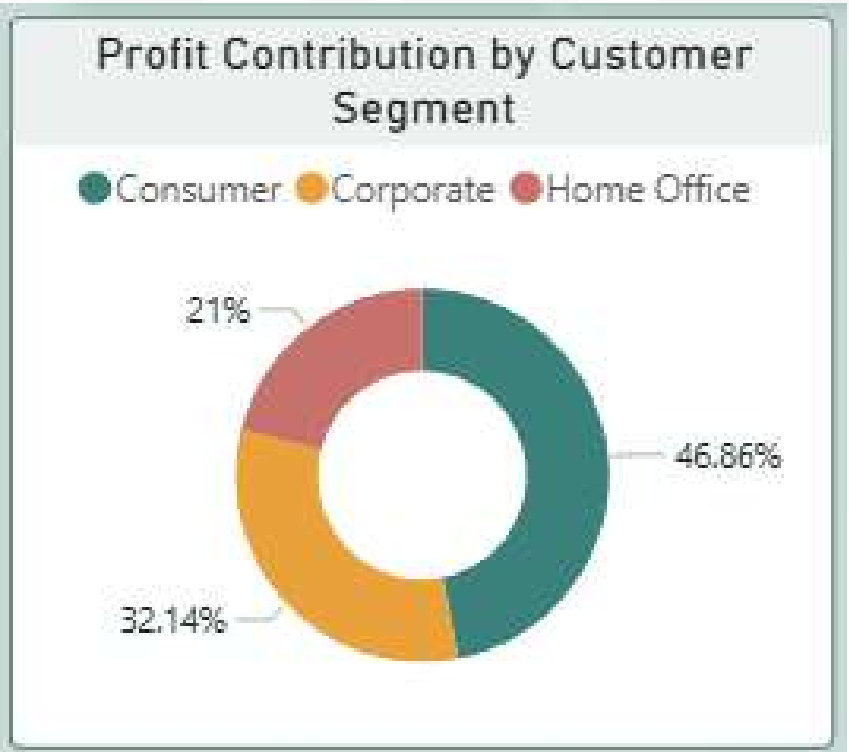
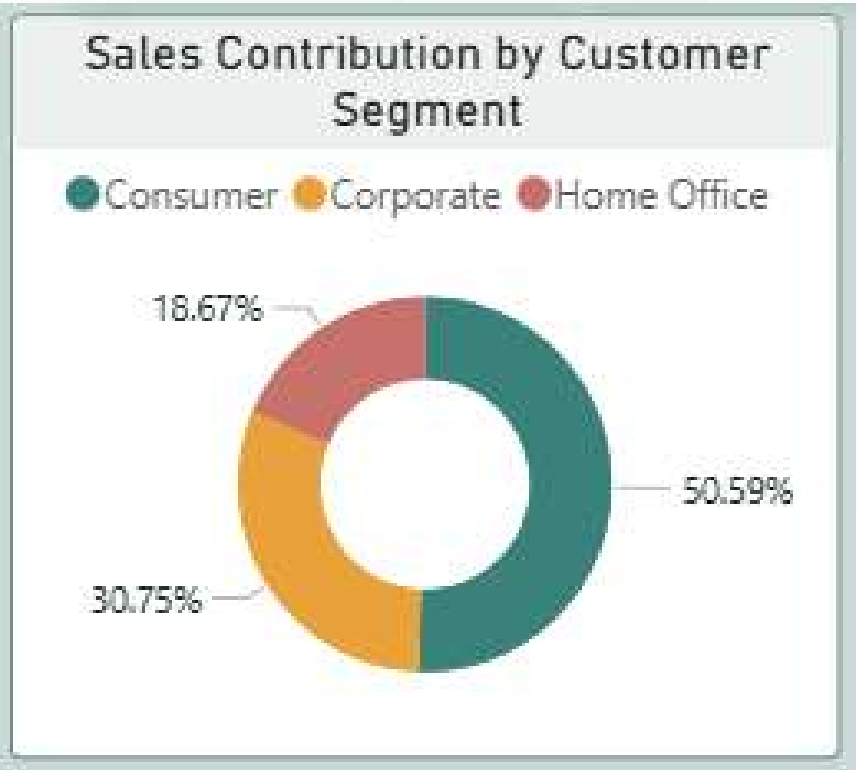


FINDING

Technology and Office Supplies generate positive profit, while Furniture records a net loss of -\$18K.

INSIGHT

Furniture is the only loss-making category despite having considerable sales volume, suggesting a pricing or cost structure problem.



FINDING

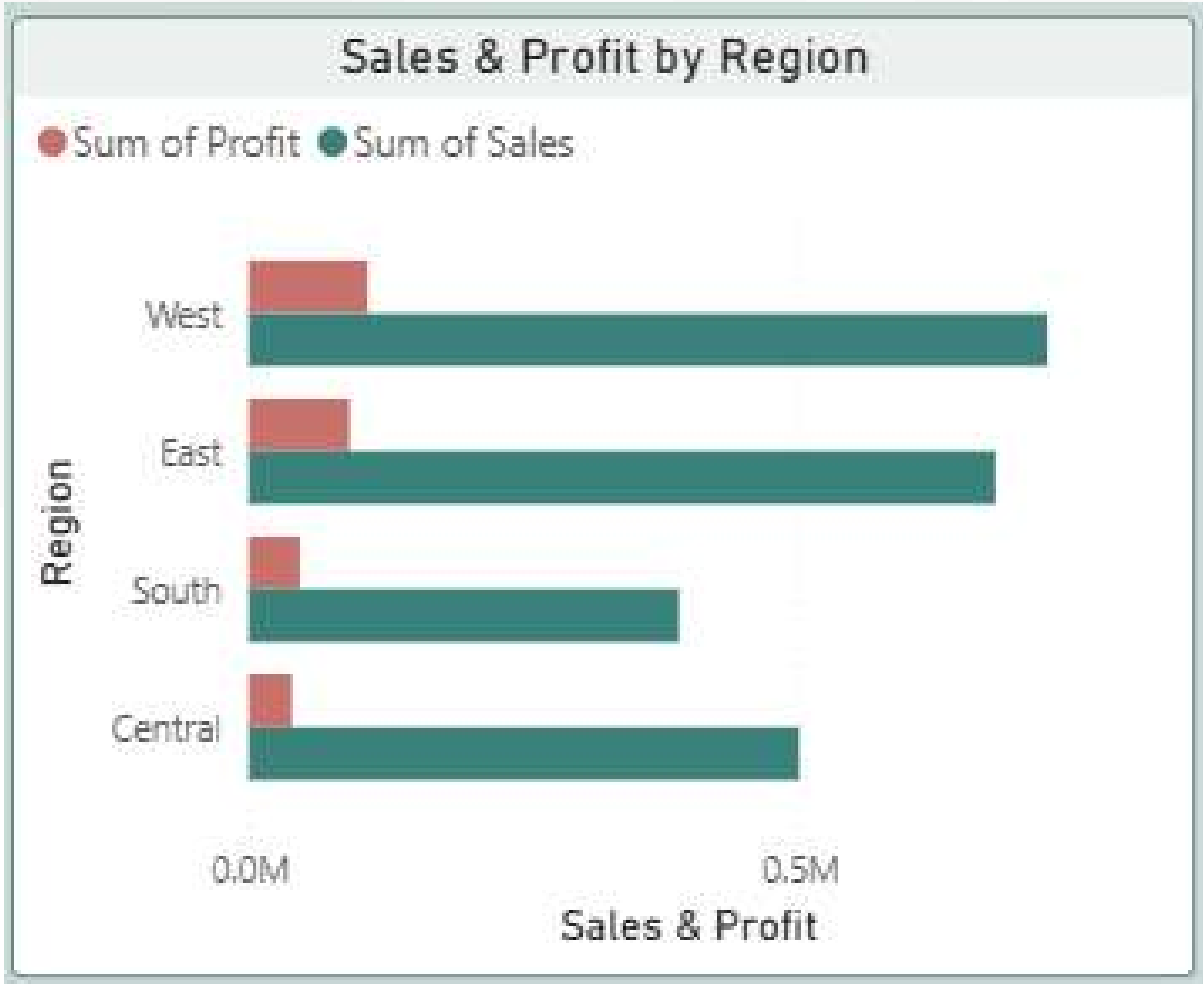
Consumer dominates revenue at 50.59%, yet contributes only 46.89% of total profit. Meanwhile Corporate contributes 30.75% of revenue but generates 32.14% of profit.

INSIGHT

Despite being the largest revenue contributor, Consumer is less profitable per dollar of sales compared to Corporate which punches above its weight in profit relative to revenue.



Interactive Sales & Profit Dashboard



FINDING



Revenue grows steadily throughout the year, peaking significantly in **Q4 (Nov-Dec)**.

INSIGHT



Despite strong Q4 sales, **profit margins narrow** during the same period, indicating heavy discounting activity during the **holiday season**.

FINDING



West region leads in both sales and profit. **Central** has moderate sales but the lowest profit.

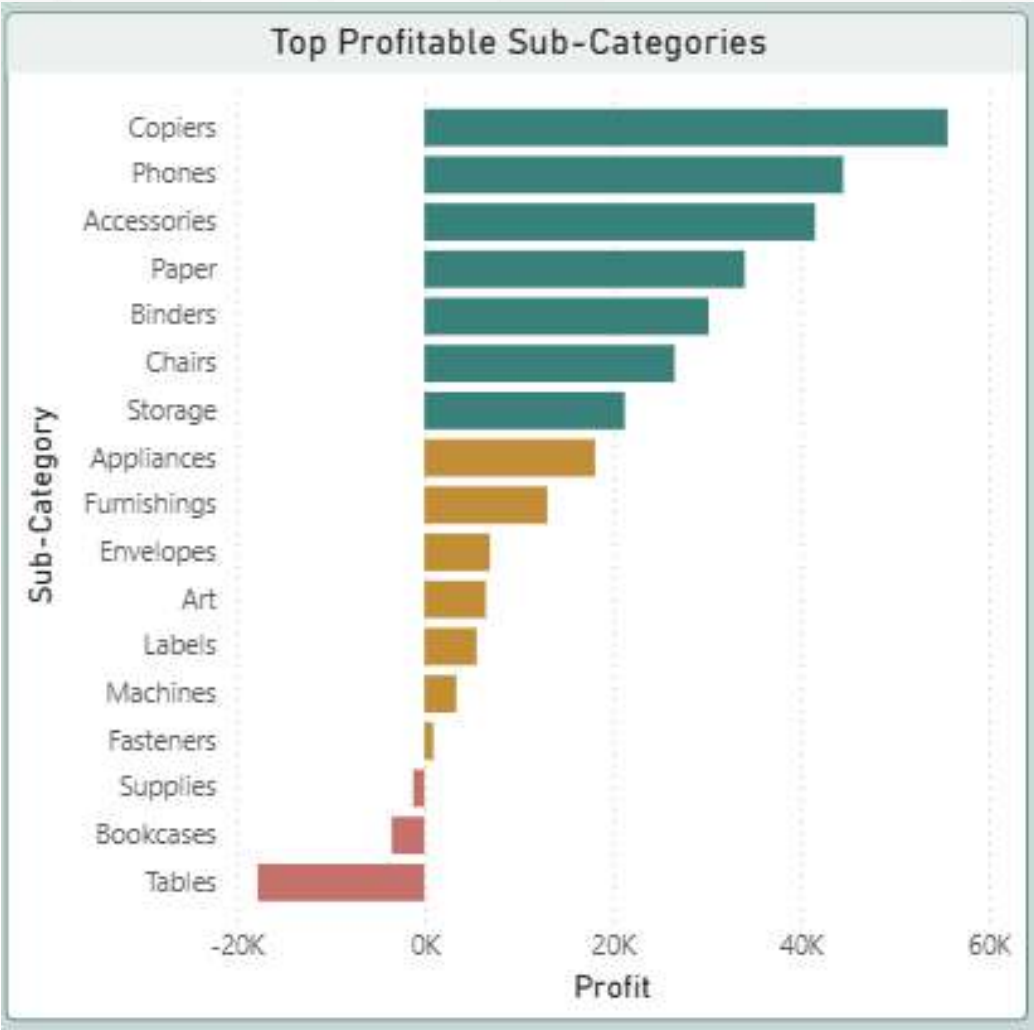
INSIGHT



Central is the only region operating at a loss, driven by the highest discount rate at 22%, significantly above other regions.



Interactive Sales & Profit Dashboard



FINDING

➤ **Copiers rank 1 with ~\$55–60K profit**, followed by Phones and Accessories. At the bottom, Tables and Bookcases record negative profit.

INSIGHT

✓ The **top 3 profitable** sub-categories all belong to Technology, **while the loss-makers Tables and Bookcases are both from Furniture**, confirming the category-level finding.

FINDING

➤ The scatter plot shows a clear **downward trend line as discount increases from 0 to 8**, profit consistently declines toward negative territory.

INSIGHT

✓ Orders with higher discounts cluster below the zero profit line, confirming that **excessive discounting is a direct driver of losses across the business**.



Business Recommendations

OPTIMIZE DISCOUNT STRATEGY

- ✓ The scatter plot confirms that higher discounts directly correlate with negative profit. **Cap discounts at 20%** and shift to targeted **promotions rather than broad price cuts.**

PRIORITIZE TECHNOLOGY CATEGORY

- ✓ **Technology leads profit** at ~\$145K with Copiers as the top sub-category. Increase inventory allocation and sales focus on this category to maximize returns.

ADDRESS FURNITURE LOSSES

- ✓ **Furniture is the only loss-making category.** Tables and Bookcases are the primary drag. **Reassess pricing, procurement costs, and discount policies** for these sub-categories.

FIX CENTRAL REGION PERFORMANCE

- ✓ **Central generates the lowest profit** despite moderate sales, driven by high discount rates. Implement stricter **discount controls** and region-specific **pricing strategies.**

LEVERAGE Q4 MORE PROFITABLY

- ✓ **Q4 drives peak** revenue but profit does not rise proportionally. **Plan inventory earlier** and **control discounting during Nov-Dec** to protect profit margins.

SHIFT FOCUS TOWARD CORPORATE SEGMENT

- ✓ **Corporate contributes 32.14%** of profit from only **30.75% of revenue** making it more **profitable per dollar of sales than Consumer.** Develop targeted offers to grow this segment further.



Conclusion



This project demonstrates a complete data analysis workflow from raw data exploration using SQL to interactive business intelligence reporting using Power BI. The analysis of 4 years of Superstore data (2014–2017) reveals that while total revenue reaches \$2.30M, profitability is significantly impacted by three key factors:

- ▶ Excessive discounting eroding profit margins
- ▶ Underperforming categories and sub-categories
- ▶ Regional and segment inefficiencies

Addressing these through data-driven decisions has the potential to meaningfully improve long-term business sustainability and profitability.

THANK YOU

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